ic_autolab round 18 (generic track): report

Protocol: PROTOCOL.md, frozen 2026-11-04T06:46:59Z at git `ddb03aeb06027907fe7d3117a7f6a8503ab5511b`; route `local`. Counted GAE only (deterministic, host independent); no wall-time claim; development loop, no validation-gate row. Loop home: `../`; ledger: `../ledger.json`.

Claim record: `PCC-20261104-13dcb4` (`python3 tools/pcc/pcc.py standing PCC-20261104-13dcb4`). Until a signed replay and recompute exist, this round's outcome is an unverified assertion [pcc:unverified].

Outcome

k class	sel	sel / inc at w = 0 [95%]	w = 0.1	w = 1	band	S sel (w = 0 / 0.1 / 1)	S inc	S rho-neg	distance sel / rho
28 **improvement**	wsum-b12-n3	0.397 [0.372, 0.424]	0.408 [0.380, 0.439]	0.471 [0.426, 0.521]	sel < inc	2.949 / 3.222 / 5.676	7.426 / 7.888 / 12.049	0.981 / 1.073 / 1.898	3.0x / 3.0x / 3.0x

Classes follow the protocol's decision rule. Distance to rho is the distance remaining, not a gate. The reasons per size:

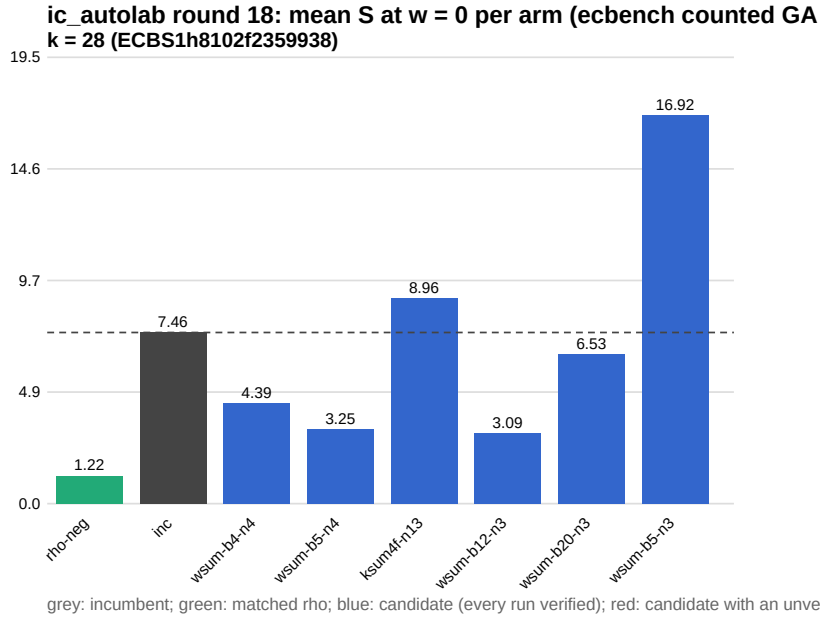
- k = 28: compare ops.status = ok, admissible = true; sel / inc = 0.3972 [0.372, 0.424] at w = 0; band: sel < inc; A/A inc-aa / inc interval [1.000, 1.000] contains 1

Train (development screen, in sample)

k = 28: session `ECBS1h8102f2359938` (run exit 0, verify exit 0, receipt sha256 `a2eab52e05dfa6c60d33b82f31f8695efba8d6bbd5ec4f8598d96894f605f136`)

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
`rho-neg`	rho	- -	`rho.negation`	32	32	1.219 / 1.334 / 2.373	matched reference
`inc`	ksum4	18 `prime-projected-abscissa:size=18,scan_cap=288`	`ksum:m=4`	32	32	7.463 / 7.932 / 12.153	incumbent
`wsum-b4-n4`	wsum-b4	4 `prime-projected-abscissa:size=4,scan_cap=64`	`wsum:coef=4,free=1`	32	32	4.389 / 5.092 / 11.421	cross-size transfer: wsum-b4-n4 is the k = 24 incumbent at N / N_a = 0.049
`wsum-b5-n4`	wsum-b5	4 `prime-projected-abscissa:size=4,scan_cap=64`	`wsum:coef=5,free=1`	32	32	3.247 / 3.593 / 6.714	cross-size transfer: wsum-b5-n3 is the k = 20 incumbent at N / N_a = 0.094
`ksum4f-n13`	ksum4f	13 `prime-projected-abscissa:size=13,scan_cap=208`	`ksum:m=4,free=1`	32	32	8.957 / 10.421 / 23.595	alternative ksum4f (best S(0) 6.675 at N = 15): N = 0.85 x 15

arm	family	N factor base	oracle	runs	verified	S (w = 0 / 0.1 / 1)	reason proposed
wsum-b12-n3	wsum-b12	3 prime-projected-abscissa:size=3,scan_cap=48	wsum:coef=12,free=1	32	32	3.094 / 3.396 / 6.107	exploration: family wsum-b12 untried at k = 28, N = 0.0001 x N_a
wsum-b20-n3	wsum-b20	3 prime-projected-abscissa:size=3,scan_cap=48	wsum:coef=20,free=1	32	32	6.533 / 6.605 / 7.253	exploration: family wsum-b20 untried at k = 28, N = 0.0001 x N_a
wsum-b5-n3	wsum-b5	3 prime-projected-abscissa:size=3,scan_cap=48	wsum:coef=5,free=1	32	32	16.923 / 20.199 / 49.690	exploration: family wsum-b5 untried at k = 28, N = 0.0001 x N_a



train screen

Selection

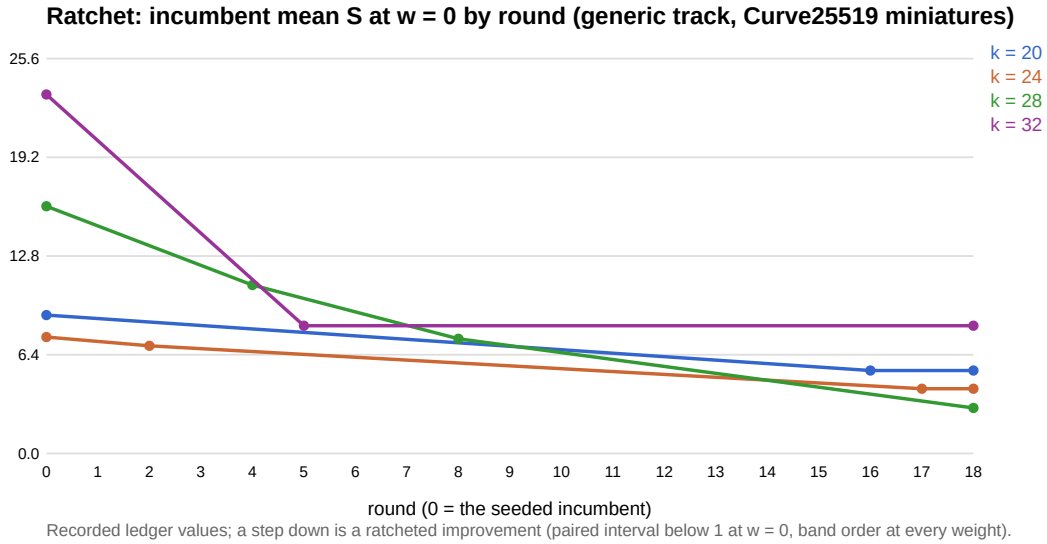
- k = 28: challenger `wsum-b12-n3` (Pareto alternative at w = 1: `wsum-b12-n3`)

Holdout sessions

k	session	run exit	verify exit	receipt sha256	levels
28	ECBS1h22b557bd4431	0	0	e44bf9dbe28a14610a5ebb0b91ef98dc430dbfe1a483c31ed36fa32930318e26	{“verified”:480}

Ledger after this round

k	incumbent	since round	S (w = 0 / 0.1 / 1)	distance to rho
20	`wsum-b5-n3` (`prime-projected-abscissa:size=3,scan_cap=48` + `wsum:coef=5,free=1`)	16	5.384 / 5.609 / 7.638	3.6x / 3.5x / 3.0x
24	`wsum-b4-n4` (`prime-projected-abscissa:size=4,scan_cap=64` + `wsum:coef=4,free=1`)	17	4.201 / 4.433 / 6.528	3.4x / 3.3x / 2.8x
28	`wsum-b12-n3` (`prime-projected-abscissa:size=3,scan_cap=48` + `wsum:coef=12,free=1`)	18	2.949 / 3.222 / 5.676	3.0x / 3.0x / 3.0x
32	`ksum4-n26` (`prime-projected-abscissa:size=26,scan_cap=416` + `ksum:m=4`)	5	8.296 / 8.811 / 13.446	9.1x / 8.8x / 7.4x



ratchet

Evidence and deviations

- ecbench binary sha256: `e7242bc192bda8c8da9e64895afe943e8aa47cfa37bd490d656e2181758e9533`; `--cpus auto`, allow_busy = true; `verify --replay 8` per session; audit receipts in `analysis/`; sessions in `sessions/` (records in the artifact store when large).
- Route: the local route is an explicitly requested development screen, not the Iso Lab EKS default; counts are deterministic and replayed, wall time is not claimed. No row of this round enters the IC leaderboard or `benchmarks/README.md` (validation gate).
- Train selection is in sample (disclosed); the holdout is fresh (new `target_seed`, excluded exposures). Failed, timed-out and regressing arms are retained above.